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5.1.1. Stacked Plot

17.51

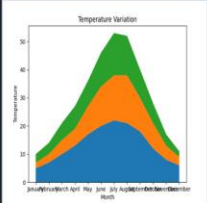
Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

Test case 1



stackedpl...

1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6 'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July',
7 'August', 'September', 'October', 'November', 'December'],
8 'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9 'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10 'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 df = pd.DataFrame(data)
15 plt.stackplot(df['Month'], df['City_A_Temperature'],
16 df['City_B_Temperature'], df['City_C_Temperature'])
17 plt.xlabel("Month")
18 plt.ylabel("Temperature")
19 plt.title("Temperature Variation")
20 plt.legend(loc='upper-left')
21 plt.show()

Average time: 0.194 s, Maximum time: 0.194 s, 1 out of 1 shown test case(s) passed

Test case 1: Expected output and Actual output both show the stacked area plot.

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5.2.1. Titanic Dataset

13.24

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions.

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.
- Title the plot as "Passenger Class Distribution".
- Label the x-axis as "Pclass" and the y-axis as "Count".

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as "Gender Distribution".

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to 8 for the histogram.
- Title the plot as "Age Distribution".

titanicDat...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the titanic dataset from the CSV file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
9
10 axes[0,0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(), color='skyblue')
11 axes[0,0].set_title("Passenger-Class Distribution")
12 axes[0,0].set_xlabel("Pclass")
13 axes[0,0].set_ylabel("Count")
14
15 axes[0,1].pie(df['Gender'].value_counts(), labels=df['Gender'].value_counts().index, autopct='%1.1f%%', colors=['lightblue', 'lightcoral'])
16 axes[0,1].set_title("Gender Distribution")
17
18 axes[1,0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
19 axes[1,0].set_title("Age Distribution")

Average time: 0.737 s, Maximum time: 0.737 s, 1 out of 1 shown test case(s) passed

Test case 1: Expected output and Actual output both show a 3x2 grid of plots: Passenger-Class Distribution, Gender Distribution, Age Distribution, and two empty plots.

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8.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use 30 bins for the histogram.

2. Set the edge color of the bars to black (k).

3. Label the x-axis as 'Age' and the y-axis as 'Frequency'.

4. Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
1	0	3	Mr. Owen Harris	male	22	0	0	21517	7.25		S	
2	1	1	Mr. John Bradley (Florence Briggs Towner)	female	38	1	0	PC 17599	71.2835	C85	C	
3	1	3	Mr. Leonard Noyes	male	20	0	0	51706	5.40		S	
4	1	1	Mr. James H. Beesley (Elizabeth Beesley)	female	35	1	0	11380	53.1	C123	S	
5	0	3	Mr. William Henry	male	30	0	0	17350	5.40		S	
6	0	3	Mr. Louis	male	0	0	0	18077	5.4583		Q	
7	0	3	Mr. Lewis	male	54	0	0	17463	51.8625		446	S
8	0	3	Mr. William Marston	male	27	1	0	14980	21.075		S	
9	1	1	Mr. Oscar W. (Ellenesth Ellenesth Berg)	female	27	0	2	147742	11.1333		S	
10	1	2	Mr. Nicholas	male	14	1	0	157736	10.8708		C	

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,Mr. Owen Harris,male,22,0,0,21517,7.25,S
2,1,1,Mr. John Bradley (Florence Briggs Towner),female,38,1,0,PC 17599,71.2835,C85,C
3,1,3,Mr. Leonard Noyes,male,20,0,0,51706,5.40,S
4,1,1,Mr. James H. Beesley (Elizabeth Beesley),female,35,1,0,11380,53.1,C123,S
5,0,3,Mr. William Henry,male,30,0,0,17350,5.40,S
6,0,3,Mr. Louis,male,0,0,0,18077,5.4583,Q
7,0,3,Mr. Lewis,male,54,0,0,17463,51.8625,446,S
8,0,3,Mr. William Marston,male,27,1,0,14980,21.075,S
9,1,1,Mr. Oscar W. (Ellenesth Ellenesth Berg),female,27,0,2,147742,11.1333,S
10,1,2,Mr. Nicholas (Adeline Armes),female,14,1,0,157736,10.8708,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Histogram...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data['Age'], bins=30, edgecolor='k')
18 plt.title('Age Distribution')
19 plt.xlabel('Age')
20 plt.ylabel('Frequency')
21 plt.show()
```

Average time: 0.156 s, Maximum time: 0.156 s, 1 out of 1 shown test case(s) passed

Test case 1

Expected output: Age Distribution histogram

Actual output: Age Distribution histogram

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8.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title "Survival Count" to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
1	0	3	Mr. Owen Harris	male	22	0	0	21517	7.25		S	
2	1	1	Mr. John Bradley (Florence Briggs Towner)	female	38	1	0	PC 17599	71.2835	C85	C	
3	1	3	Mr. Leonard Noyes	male	20	0	0	51706	5.40		S	
4	1	1	Mr. James H. Beesley (Elizabeth Beesley)	female	35	1	0	11380	53.1	C123	S	
5	0	3	Mr. William Henry	male	30	0	0	17350	5.40		S	
6	0	3	Mr. Louis	male	0	0	0	18077	5.4583		Q	
7	0	3	Mr. Lewis	male	54	0	0	17463	51.8625		446	S
8	0	3	Mr. William Marston	male	27	1	0	14980	21.075		S	
9	1	1	Mr. Oscar W. (Ellenesth Ellenesth Berg)	female	27	0	2	147742	11.1333		S	
10	1	2	Mr. Nicholas (Adeline Armes)	female	14	1	0	157736	10.8708		C	

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,Mr. Owen Harris,male,22,0,0,21517,7.25,S
2,1,1,Mr. John Bradley (Florence Briggs Towner),female,38,1,0,PC 17599,71.2835,C85,C
3,1,3,Mr. Leonard Noyes,male,20,0,0,51706,5.40,S
4,1,1,Mr. James H. Beesley (Elizabeth Beesley),female,35,1,0,11380,53.1,C123,S
5,0,3,Mr. William Henry,male,30,0,0,17350,5.40,S
6,0,3,Mr. Louis,male,0,0,0,18077,5.4583,Q
7,0,3,Mr. Lewis,male,54,0,0,17463,51.8625,446,S
8,0,3,Mr. William Marston,male,27,1,0,14980,21.075,S
9,1,1,Mr. Oscar W. (Ellenesth Ellenesth Berg),female,27,0,2,147742,11.1333,S
10,1,2,Mr. Nicholas (Adeline Armes),female,14,1,0,157736,10.8708,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar-Plot-for-Survival-Rate
17 survival_counts = data['Survived'].value_counts()
18 survival_counts.plot(kind='bar')
19 plt.title('Survival Count')
20 plt.xlabel('Survived')
21 plt.ylabel('Count')
```

Average time: 0.178 s, Maximum time: 0.178 s, 1 out of 1 shown test case(s) passed

Test case 1

Expected output: Survival Count bar chart

Actual output: Survival Count bar chart

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8.2.4. Bar Plot for Survival by Gender

0.25s

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Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a **stacked bar chart** to display the survival counts.
- Add the title "Survival by Gender" to the chart.
- Label the x-axis as 'Gender' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Thomas Bruns	male	22	0	0	21517	7.25		S
2	1	1	Mr. John Bradley	female	18	1	0	PC 17599	71.2835	C85	C
3	0	3	Mr. James Watson	male	30	0	0	51709	53.1	C123	S
4	1	1	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
5	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
6	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
7	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
8	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
9	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
10	1	1	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
11	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
12	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
13	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
14	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
15	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
16	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
17	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
18	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
19	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
20	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
21	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, Mr. Thomas Bruns, male, 22, 0, 0, 21517, 7.25, , S
2, 1, 1, Mr. John Bradley, female, 18, 1, 0, PC 17599, 71.2835, C85, C
3, 0, 3, Mr. James Watson, male, 30, 0, 0, 51709, 53.1, C123, S
4, 1, 1, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
5, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
6, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
7, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
8, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
9, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
10, 1, 1, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
11, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
12, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
13, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
14, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
15, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
16, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
17, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
18, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
19, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
20, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
21, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotOf...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16# Write your code here for Bar Plot for Survival by Gender
17survival_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
18survival_counts.plot(kind='bar', stacked=True)
19plt.title("Survival by Gender")
20plt.xlabel("Gender")
21plt.ylabel("Count")

Average time0.173 s
Maximum time0.173 s
1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

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Test Cases

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8.2.5. Bar Plot for Survival by Pclass

0.51s

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Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the 'Pclass' column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using `value_counts()`.
- Use a **stacked bar chart** to display the survival counts.
- Add the title "Survival by Pclass" to the chart.
- Label the x-axis as 'Pclass' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Thomas Bruns	male	22	0	0	21517	7.25		S
2	1	1	Mr. John Bradley	female	18	1	0	PC 17599	71.2835	C85	C
3	0	3	Mr. James Watson	male	30	0	0	51709	53.1	C123	S
4	1	1	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
5	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
6	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
7	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
8	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
9	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
10	1	1	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
11	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
12	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
13	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
14	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
15	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
16	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
17	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
18	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
19	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
20	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S
21	0	3	Mr. William Miller	male	35	0	0	11380	53.1	C123	S

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, Mr. Thomas Bruns, male, 22, 0, 0, 21517, 7.25, , S
2, 1, 1, Mr. John Bradley, female, 18, 1, 0, PC 17599, 71.2835, C85, C
3, 0, 3, Mr. James Watson, male, 30, 0, 0, 51709, 53.1, C123, S
4, 1, 1, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
5, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
6, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
7, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
8, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
9, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
10, 1, 1, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
11, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
12, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
13, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
14, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
15, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
16, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
17, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
18, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
19, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
20, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
21, 0, 3, Mr. William Miller, male, 35, 0, 0, 11380, 53.1, C123, S
```

Note:

- Refer to the visible test case for better reference.
- Ensure you use the `groupby()` function with `value_counts()` to count the survivors and non-survivors for each Pclass.
- Do not manually use `size()` or `unstack()` without `value_counts()`. Use the `value_counts()` method for counting survival status directly.

Sample Test Cases

BarPlotOf...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16# Write your code here for Bar Plot for Survival by Pclass
17survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack()
18survival_counts.plot(kind='bar', stacked=True)
19plt.title("Survival by Pclass")
20plt.xlabel("Pclass")
21plt.ylabel("Count")

Average time0.203 s
Maximum time0.203 s
1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Terminal

Test Cases

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5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

- Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
- Set the chart type to 'bar' and make it stacked.
- Label the title "**Survival by Embarked**" to the chart.
- Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
- Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21373, 71	5.69	D	S
2	1	1	Gale, Mrs. Jane Seabird	female	38	1	0	PC 17599, 71	26.33	C85,C	S
3	1	3	Nicolson, Miss. Helen	female	26	0	0	STON/O 3101282, 7	9.67		S
4	1	3	McCarthy, Mrs. Thomas	female	25	0	0	W/14900, 21	8.67		S
5	0	3	O'Brien, Mr. William Henry	male	35	0	0	W/15003, 53	1.52	C223,S	S
6	0	3	Harris, Mr. James	male	40	3	0	W/15009, 9	8.51		S
7	0	3	Harper, Mr. James	male	38	0	0	W/15017, 8	4.58		Q
8	0	3	McCarthy, Mr. Timothy	male	54	0	0	W/17803, 51	8.02	E66,S	S
9	0	3	McCarthy, Master. Patrick	male	7	1	1	W/14900, 21	8.67		S
10	1	3	Harrison, Mrs. Susan W. (Misses Harrison)	female	27	0	0	W/14780, 11	13.33		S
11	0	3	Nelson, Mrs. Nicholas (Nelson Army)	female	34	1	0	W/17736, 38	8.08		C

Sample Data:

```

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,Braund,Mr.Owen Harris,male,22,1,0,A/5 21373,71,5.69,D,S
2,1,1,Gale,Mrs.Jane Seabird,female,38,1,0,PC 17599,71,26.33,C85,C
3,1,3,Nicolson,Miss.Helen,female,26,0,0,STON/O 3101282,7,9.67,S
4,1,3,McCarthy,Mrs.Thomas,female,25,0,0,W/14900,21,8.67,S
5,0,3,O'Brien,Mr.William Henry,male,35,0,0,W/15003,53,1.52,C223,S
6,0,3,Harris,Mr.James,male,40,3,0,W/15009,9,8.51,S
7,0,3,Harper,Mr.James,male,38,0,0,W/15017,8,4.58,Q
8,0,3,McCarthy,Mr.Timothy,male,54,0,0,W/17803,51,8.02,E66,S
9,0,3,McCarthy,Master.Patrick,male,7,1,1,W/14900,21,8.67,S
10,1,3,Harrison,Mrs.Susan W.(Misses Harrison),female,27,0,0,W/14780,11,13.33,S
11,0,3,Nelson,Mrs.Nicholas(Nelson Army),female,34,1,0,W/17736,38,8.08,C
    
```

Note: Refer to the visible test case for better reference.

BarPlotOf...

```

import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'] = data['Age'].median(), inplace=True
data['Embarked'] = data['Embarked'].mode()[0], inplace=True
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar-Plot for Survival by Embarked
survival_counts = data.groupby("Embarked_Q")["Survived"].value_counts().unstack()
survival_counts.plot(kind='bar', stacked=True)
plt.title("Survival by Embarked")
plt.xlabel("Embarked")
plt.ylabel("Count")
    
```

Average time: 0.225 s Maximum time: 0.225 ms 1 out of 1 shown test case(s) passed

Test case 1: PASS & DEBUG

Expected output

Actual output

Terminal Test cases

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5.2.8. Box Plot for Age by Survived

0125

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).

2. Set the title of the plot to **"Age by Survival"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **"Survived"** and the y-axis as **"Age"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Harris	male	22	0	0	211717.26	5		
2	1	1	Mr. John Bradley (Ellenor Briggs) Taylor	female	38	1	0	PC 17599,71-2835,C85,C			
3	1	3	Mr. Robert Williams	male	30	0	0	STON/O 3182829,7-000,5			
4	1	1	Mr. James Cameron	male	35	1	0	113803,53-1,C123,5			
5	0	3	Mr. William Henry	male	30	0	0	173450,8-00,5			
6	0	3	Mr. James	male	0	0	0	18077,8-4583,Q			
7	0	3	Mr. James	male	54	0	0	17469,31-0803,446,5			
8	0	3	Mr. William Hunter	male	27	1	1	148969,21-070,5			
9	1	1	Mr. James	male	0	0	0	1113803,53-1,C123,5			
10	1	2	Mr. William Hunter	male	27	0	0	147742,11-133,5			
11	1	2	Mr. William Hunter	male	14	1	0	157736,10-0700,C			

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1, 0, 3, Mr. Owen Harris, male, 22, 0, 0, 211717.26, 5, , ,

2, 1, 1, Mr. John Bradley (Ellenor Briggs) Taylor, female, 38, 1, 0, PC 17599,71-2835,C85,C, , ,

3, 1, 3, Mr. Robert Williams, male, 30, 0, 0, STON/O 3182829,7-000,5, , ,

4, 1, 1, Mr. James Cameron, male, 35, 1, 0, 113803,53-1,C123,5, , ,

5, 0, 3, Mr. William Henry, male, 30, 0, 0, 173450,8-00,5, , ,

6, 0, 3, Mr. James, male, 0, 0, 0, 18077,8-4583,Q, , ,

7, 0, 3, Mr. James, male, 54, 0, 0, 17469,31-0803,446,5, , ,

8, 0, 3, Mr. William Hunter, male, 27, 1, 1, 148969,21-070,5, , ,

9, 1, 1, Mr. James, male, 0, 0, 0, 1113803,53-1,C123,5, , ,

10, 1, 2, Mr. William Hunter, male, 27, 0, 0, 147742,11-133,5, , ,

11, 1, 2, Mr. William Hunter, male, 14, 1, 0, 157736,10-0700,C, , ,

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv("Titanic-Dataset.csv")

6

7# Data Cleaning

8data["Age"].fillna(data["Age"].median(), inplace=True)

9data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)

10data.drop("Cabin", axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=["Embarked"], drop_first=True)

15

16# Write your code here for Box Plot for Age by Survived

17data.boxplot(column="Age", by="Survived")

18plt.title("Age by Survival")

19plt.suptitle("")

20plt.xlabel("Survived")

21plt.ylabel("Age")

Average time0.203 sMaximum time0.203 s1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

TerminalTest Cases

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5.2.9. Box Plot for Fare by Pclass

0125

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.

2. Set the title of the plot to **"Fare by Pclass"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **"Pclass"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Harris	male	22	0	0	211717.26	5		
2	1	1	Mr. John Bradley (Ellenor Briggs) Taylor	female	38	1	0	PC 17599,71-2835,C85,C			
3	1	3	Mr. Robert Williams	male	30	0	0	STON/O 3182829,7-000,5			
4	1	1	Mr. James Cameron	male	35	1	0	113803,53-1,C123,5			
5	0	3	Mr. William Henry	male	30	0	0	173450,8-00,5			
6	0	3	Mr. James	male	0	0	0	18077,8-4583,Q			
7	0	3	Mr. James	male	54	0	0	17469,31-0803,446,5			
8	0	3	Mr. William Hunter	male	27	1	1	148969,21-070,5			
9	1	1	Mr. James	male	0	0	0	1113803,53-1,C123,5			
10	1	2	Mr. William Hunter	male	27	0	0	147742,11-133,5			
11	1	2	Mr. William Hunter	male	14	1	0	157736,10-0700,C			

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1, 0, 3, Mr. Owen Harris, male, 22, 0, 0, 211717.26, 5, , ,

2, 1, 1, Mr. John Bradley (Ellenor Briggs) Taylor, female, 38, 1, 0, PC 17599,71-2835,C85,C, , ,

3, 1, 3, Mr. Robert Williams, male, 30, 0, 0, STON/O 3182829,7-000,5, , ,

4, 1, 1, Mr. James Cameron, male, 35, 1, 0, 113803,53-1,C123,5, , ,

5, 0, 3, Mr. William Henry, male, 30, 0, 0, 173450,8-00,5, , ,

6, 0, 3, Mr. James, male, 0, 0, 0, 18077,8-4583,Q, , ,

7, 0, 3, Mr. James, male, 54, 0, 0, 17469,31-0803,446,5, , ,

8, 0, 3, Mr. William Hunter, male, 27, 1, 1, 148969,21-070,5, , ,

9, 1, 1, Mr. James, male, 0, 0, 0, 1113803,53-1,C123,5, , ,

10, 1, 2, Mr. William Hunter, male, 27, 0, 0, 147742,11-133,5, , ,

11, 1, 2, Mr. William Hunter, male, 14, 1, 0, 157736,10-0700,C, , ,

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv("Titanic-Dataset.csv")

6

7# Data Cleaning

8data["Age"].fillna(data["Age"].median(), inplace=True)

9data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)

10data.drop("Cabin", axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=["Embarked"], drop_first=True)

15

16# Write your code here for Box Plot for Fare by Pclass

17data.boxplot(column="Fare", by="Pclass")

18plt.title("Fare by Pclass")

19plt.suptitle("")

20plt.xlabel("Pclass")

21plt.ylabel("Fare")

Average time0.173 sMaximum time0.173 s1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

TerminalTest Cases

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8.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Set the title of the plot to **"Age vs. Fare"**

3. Label the x-axis as **"Age"** and the y-axis as **"Fare"**

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Henry	male	22	0	0	A/5 21171,726	5		
2	1	1	Mr. John Bradley	male	30	1	0	PC 17599,71-2635,C85,C	53.1		
3	1	3	Mr. Otto	male	20	0	0	STON/O 3102629,7	20.5		
4	1	1	Mr. Henry	male	30	1	0	A/5 21100,53-1,C123,S	53.1		
5	0	3	Mr. William Henry	male	35	0	0	173450,8-05,S	53.1		
6	0	3	Mr. James	male	0	0	0	130777,8-4583,Q	53.1		
7	0	3	Mr. James	male	54	0	0	17460,23-0023,446,S	53.1		
8	0	3	Mr. James	male	21	1	1	134900,21-079,S	53.1		
9	1	1	Mr. Oscar W. (Elizabeth Vilhelmina Berg)	female	27	0	2	347742,11-1333,S	53.1		
10	1	2	Mr. Michael (Adelle Agnes)	female	14	1	0	137736,10-0700,C	53.1		

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1, 0, 3, Mr. Owen Henry, male, 22, 0, 0, A/5 21171,726, 5, , ,

2, 1, 1, Mr. John Bradley, male, 30, 1, 0, PC 17599,71-2635,C85,C, 53.1, , ,

3, 1, 3, Mr. Otto, male, 20, 0, 0, STON/O 3102629,7, 20.5, , ,

4, 1, 1, Mr. Henry, male, 30, 1, 0, A/5 21100,53-1,C123,S, 53.1, , ,

5, 0, 3, Mr. William Henry, male, 35, 0, 0, 173450,8-05,S, 53.1, , ,

6, 0, 3, Mr. James, male, 0, 0, 0, 130777,8-4583,Q, 53.1, , ,

7, 0, 3, Mr. James, male, 54, 0, 0, 17460,23-0023,446,S, 53.1, , ,

8, 0, 3, Mr. James, male, 21, 1, 1, 134900,21-079,S, 53.1, , ,

9, 1, 1, Mr. Oscar W. (Elizabeth Vilhelmina Berg), female, 27, 0, 2, 347742,11-1333,S, 53.1, , ,

10, 1, 2, Mr. Michael (Adelle Agnes), female, 14, 1, 0, 137736,10-0700,C, 53.1, , ,

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareS...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv("Titanic-Dataset.csv")

6

7 # Data Cleaning

8 data["Age"].fillna(data["Age"].median(), inplace=True)

9 data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)

10 data.drop("Cabin", axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=["Embarked"], drop_first=True)

15

16 # Write your code here for Box Plot for Fare by Pclass

17 plt.scatter(data["Age"], data["Fare"])

18 plt.title("Age vs. Fare")

19 plt.xlabel("Age")

20 plt.ylabel("Fare")

21 plt.show()

Average time 0.144 s Maximum time 0.144 s 1 out of 1 shown test case(s) passed

Test case 1 Expected output Actual output

Age vs. Fare

Terminal Test Cases

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8.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.

2. Color the points based on the **Survived** column. Red for passengers who did not survive (**Survived = 0**). Blue for passengers who survived (**Survived = 1**).

3. Set the title of the plot to **"Age vs. Fare by Survival"**

4. Label the x-axis as **"Age"** and the y-axis as **"Fare"**

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Henry	male	22	0	0	A/5 21171,726	5		
2	1	1	Mr. John Bradley	male	30	1	0	PC 17599,71-2635,C85,C	53.1		
3	1	3	Mr. Otto	male	20	0	0	STON/O 3102629,7	20.5		
4	1	1	Mr. Henry	male	30	1	0	A/5 21100,53-1,C123,S	53.1		
5	0	3	Mr. William Henry	male	35	0	0	173450,8-05,S	53.1		
6	0	3	Mr. James	male	0	0	0	130777,8-4583,Q	53.1		
7	0	3	Mr. James	male	54	0	0	17460,23-0023,446,S	53.1		
8	0	3	Mr. James	male	21	1	1	134900,21-079,S	53.1		
9	1	1	Mr. Oscar W. (Elizabeth Vilhelmina Berg)	female	27	0	2	347742,11-1333,S	53.1		
10	1	2	Mr. Michael (Adelle Agnes)	female	14	1	0	137736,10-0700,C	53.1		

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1, 0, 3, Mr. Owen Henry, male, 22, 0, 0, A/5 21171,726, 5, , ,

2, 1, 1, Mr. John Bradley, male, 30, 1, 0, PC 17599,71-2635,C85,C, 53.1, , ,

3, 1, 3, Mr. Otto, male, 20, 0, 0, STON/O 3102629,7, 20.5, , ,

4, 1, 1, Mr. Henry, male, 30, 1, 0, A/5 21100,53-1,C123,S, 53.1, , ,

5, 0, 3, Mr. William Henry, male, 35, 0, 0, 173450,8-05,S, 53.1, , ,

6, 0, 3, Mr. James, male, 0, 0, 0, 130777,8-4583,Q, 53.1, , ,

7, 0, 3, Mr. James, male, 54, 0, 0, 17460,23-0023,446,S, 53.1, , ,

8, 0, 3, Mr. James, male, 21, 1, 1, 134900,21-079,S, 53.1, , ,

9, 1, 1, Mr. Oscar W. (Elizabeth Vilhelmina Berg), female, 27, 0, 2, 347742,11-1333,S, 53.1, , ,

10, 1, 2, Mr. Michael (Adelle Agnes), female, 14, 1, 0, 137736,10-0700,C, 53.1, , ,

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFareS...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv("Titanic-Dataset.csv")

6

7 # Data Cleaning

8 data["Age"].fillna(data["Age"].median(), inplace=True)

9 data["Embarked"].fillna(data["Embarked"].mode()[0], inplace=True)

10 data.drop("Cabin", axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data["Sex"] = data["Sex"].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=["Embarked"], drop_first=True)

15

16 # Write your code here for Scatter Plot for Age vs. Fare by Survived

17 colors = data["Survived"].map({0: 'Red', 1: 'Blue'})

18 plt.scatter(data["Age"], data["Fare"], c=colors)

19 plt.title("Age vs. Fare by Survival")

20 plt.xlabel("Age")

21 plt.ylabel("Fare")

Average time 0.162 s Maximum time 0.162 s 1 out of 1 shown test case(s) passed

Test case 1 Expected output Actual output

Age vs. Fare by Survival

Terminal Test Cases